

PSP Oracle Exit - Monolith Database Migration Cutover Runbook (Postgres)

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Goal

- Validate Deployment, App and Database Re-architecture changes for Postgres Hash Migration.
- Understand and learn the system behaviour functional/non functional in the Postgres Hash database.
- Ensure the system's performance, including query speed and overall responsiveness, remains unaffected or improved post-migration.
- Continuously monitor the system post-migration for stability, keeping an eye out for unexpected crashes, errors, or any other issues indicating potential migration problems.

Cutover Summary

Status

Environment	Status
Production	PLANNED

Cutover Progress

JIRA	PSP-28546 - Getting issue details... STATUS
Start Time	03 Nov 2023 11 AM IST
Slack Channel	#psp-monolith-postgres-hash-prod-cutover

Snoozing A2i

S. No	Application	A2I Maintenance URL
1	DT-Payroll-Svc	https://devportal.intuit.com/app/dp/resource/6847740825712235356/incidentManagement
2	QBDT-Payroll-Service	https://devportal.intuit.com/app/dp/resource/7999536430416583774/incidentManagement
3	CEP	https://devportal.intuit.com/app/dp/resource/6343337667453040796/incidentManagement
4	Case 360	https://devportal.intuit.com/app/dp/resource/1587536460949765378/incidentManagement
5	EWS	https://devportal.intuit.com/app/dp/resource/8649180676662812116/incidentManagement
6	PTC	https://devportal.intuit.com/app/dp/resource/3315792817950450305/incidentManagement
7	QBDT Payroll Entitlement	https://devportal.intuit.com/app/dp/resource/6990730113881900139/incidentManagement (Not onboarded to A2i)
8	Workforce	https://devportal.intuit.com/app/dp/resource/2956630747670426490/incidentManagement
9	BRM	https://devportal.intuit.com/app/dp/resource/3893379470159715175/incidentManagement (Not onboarded to A2i)
10	Desktop Pay API	https://devportal.intuit.com/app/dp/resource/1730525749114265769/incidentManagement
11	Payroll DM	https://devportal.intuit.com/app/dp/resource/8866479358728757267/incidentManagement (Not onboarded to A2i since its not behind gateway)
12	Pubsub	https://devportal.intuit.com/app/dp/resource/651913638447486453/incidentManagement (Not onboarded to A2i since its not behind gateway)
13	EWS Service	https://devportal.intuit.com/app/dp/resource/4901059886671999792/incidentManagement
14	EWS-Rest-IKS Service	https://devportal.intuit.com/app/dp/resource/6270154173504901366/incidentManagement

Downtime Details

App	Cutover (PST) 03-Nov-2023			Emergency Rollback (PST)		
	Total Downtime : 1hr 18min					
	Downtime Start	Downtime End	Total Time	Downtime Start	Downtime End	Actual
PSP	12:40 AM	01:40 AM	60min	NA	NA	NA
PAPI	12:36 AM	01:50 AM	1hr 14min	NA	NA	NA
Pubsub	12:36 AM	01:48 AM	1hr 12min	NA	NA	NA
RJF	12:36 AM	01:50 AM	1hr 14min	NA	NA	NA
Payroll DM	12:36 AM	01:49 AM	1hr 13min	NA	NA	NA

Downtime Summary

Downtime	Estimated	Actual	Cutover/Emergency Rollback Window		Comments
			Start Time (PST)	End Time (PST)	

Cutover	105 Mins				Time Split Expected 1 hour 15 mins of Deploying all apps at full scale. 30 Mins for database side steps - Database Replication Lag is 0. - Stopping A B and creating B A replication tasks or vice versa in case of rollback.
Planned Rollback (Not Applicable for Cutover)	120 Mins				Time Split Expected - Database Replication Lag is 0. 15 Mins for Sanity.
Emergency Rollback (In case of issues during the Cutover window)	170 Mins				Time Split Expected 120 Mins for Planned Rollback 30 Mins Debugging and Replication Lag to settle down. 15 Mins of additional time for Sanity.

Learnings/Enhancements

S No	Learning	Action Item	Owner	Priority	Status	Comments
1						
2						
3						

Important Resources

Dev Portal

Application	Dev Portal Link
PSP	PSP Dev Portal
PAPI	PAPI Dev Portal
PubSub	Pubsub Dev Portal
RJF	RJF Dev Portal
PayrollIDM	PayrollIDM Dev Portal

JSS

Application	JSS Link
PSP Batch Jobs	PSP JSS View
RJF Reports	RJF JSS View

Splunk

Type	Link
Application	Splunk Application Production
Gateway	Splunk Gateway Production

ArgoCd

Environment	Application	Dev Portal Link
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Production	PSP	Production PSP ArgoCd
	PAPI	Production PAPI ArgoCd
	Pubsub	Production Pubsub ArgoCd
	RJF	Production RJF ArgoCd
	PayrollDM	Production PayrollDM ArgoCd

Notification Email Addresses

i SBSEG-PD QBDT PSP AWS <SBSEG-PDQBDTSPAWS@intuit.com>; SBG PSP Services Decom Scrum <SBGSPServicesDecompScrum@intuit.com>; SBSEG SBG Payroll PSP Informed <SBGPayrollPSPInformed@intuit.com>; SBSEG-Iqbal-Extended <SBSEG-PDIqbalExtended@intuit.com>; SBSEG-PD DTPayrollExtended <SBSEG-PDDTPayrollExtended@intuit.com>; SBSEG-PD DTPayroll <SBSEG-PDDTPayroll@intuit.com> Kaura, Varun <Varun_Kaura@intuit.com>; Narlagalla, Praveen Goud <PraveenGoud_Narlagalla@intuit.com>; Swain, Amrit <Amrit_Swain@intuit.com>; Johnson, Anu <Anu_Johnson@intuit.com>, anuradha_kumar_i@intuit.com, rohan_rasam@intuit.com, narendraprasad_nanna@intuit.com, subulakshmi_nagarajan@intuit.com, rajeevkumar_singh@intuit.com, ketan_popat@intuit.com, Pahuja, Hitesh <Hitesh_Pahuja@intuit.com>, Huang, Alex <Alex_Huang@intuit.com>

Change Requests

Application	Cutover			Emergency Rollback		
	Change Request	Deployment Image	Status	Change Request	Deployment Image	Status
PSP	CHG2607081	master-3b851e3	OPEN	NA	NA	NA
PAPI	CHG2607101	master-bb6d80e	OPEN	NA	NA	NA
Pubsub	CHG2607119	master-f4b731b	OPEN	NA	NA	NA
RJF	CHG2607126	master-25ad235	OPEN	NA	NA	NA
PayrollDM	CHG2607175	master-dd1b894	OPEN	NA	NA	NA

Config Map Pull requests

Cutover Pull Requests

Application	Pull Requests
PSP	Link
PAPI	Link
Pubsub	Link
RJF	NA

Rollback Pull Requests

Application	Pull Requests
PSP	
PAPI	
Pubsub	
RJF	

Runbook

Pre Cutover Steps

S No	Task	Steps	Due Date	Status	Owner								
1	Partner Sign-off.	Refer NotificationEmailAddresses Section. Partner Contacts : Partner Team Partner Signoff	31 Oct 2023	DONE	Nagarajan, Subulakshmi Nanna, Narendra Prasad								
2	Inform partner teams slack channel.	Ask the stake holders to be prepared for the downtime. Inform following Slack Channels. #payroll-pd-releases, #ioc, #psp-rtb-priority, #sbg-payroll-prod, #mse-commerce-support, #cx3-production Partner teams communication ioc-signoff sbg-payroll-prod-signoff psp-rtb-priority-signoff payroll-pd-releases-signoff psp-inform mse-signoff cx3-production-signoff	31 Oct 2023	DONE	N, Ramasubramani								
3	Send Release Freeze Email to Dev Team.	No Releases in PSP / PAPI / Pubsub / RJF / PayrollDM allowed in this interval.	31 Oct 2023	DONE	Bhat, Chethan								
4	Care Sign-off	SAP can be validated in Staging Environment. Partner Signoff Slack Channel : #psp-sap-validation	31 Oct 2023	DONE	Sur, Koustav								
5	Inform Leadership Team on the Risk & Downtime	Inform Leadership Team on the Risk & Downtime.	31 Oct 2023	DONE	Nasim, Syed Iqbal								
6	Prepare QBDT Production Setup.	Prepare QBDT Production Setup. QBDT : 106924271 Company: Right Networks	31 Oct 2023	DONE	K, Srividya Hoolimath, Praveenkumar								
7	Schema object count check	Count of objects are matching except known ones for partitioning.	01 Nov 2023	DONE	Choubey, Mayank								
8	Time for Validating Latest data during the Cutover window	Rajoria, Himanshu to update expected numbers and veridata url and job names. https://goldengate-1.sbg-psp-prod.a.intuit.com/veridata/login.jsf PSP_O2P_last_6_hrs_B_C2 <table><tr><th>Activity</th><th>Job Name</th><th>Expected</th><th>Actual</th></tr><tr><td>Tiptoe</td><td></td><td>10 mins</td><td></td></tr></table>	Activity	Job Name	Expected	Actual	Tiptoe		10 mins		03 Nov 2023	DONE	Rajoria, Himanshu
Activity	Job Name	Expected	Actual										
Tiptoe		10 mins											
9	Decide strategy for Monthly & Quarterly Jobs.	Decide strategy for Monthly & Quarterly Jobs.	01 Nov 2023	DONE	Swain, Amrit								
10	Make sure there is no Splunk/Git/IBP /MSaaS Maintenance during the Cutover window.	- Splunk Signoff : splunk-signoff - MSaaS Signoff - msaas-signoff - IBP Signoff - ibp-signoff - Git Signoff - git-signoff Get the contacts of MSaaS team in case if we need assistance during Cutover if required.	01 Nov 2023	DONE	Bansal, Jatin								
11	Capture previous traffic for each app.	- Update AppsHealth section with the last week traffic. - Post the traffic and health numbers in Cutover channel. Window : 26th Oct 11:30PM PST - 27th Oct 05:30AM PST	01 Nov 2023	DONE	Singh, Sanket Unknown User (akumari3) Goswami, Harshita Singhal, Vikas								

12	Make sure the last Master branch deployed did not have any deployment errors.	Look for errors in this dashboard <table><tr><th>App</th><th>Dashboard</th><th>Status</th></tr><tr><td>PSP</td><td>PSP Unknown Errors</td><td>NO ERRORS</td></tr><tr><td>PAPI</td><td>PAPI Unknown Errors</td><td>NO ERRORS</td></tr><tr><td>Pubsub</td><td>PubSub Unknown Errors</td><td>NO ERRORS</td></tr><tr><td>RJF</td><td>RJF Unknown Errors</td><td>NO ERRORS</td></tr><tr><td>PayrollDM</td><td>PayrollDM Unknown Errors</td><td>NO ERRORS</td></tr></table>	App	Dashboard	Status	PSP	PSP Unknown Errors	NO ERRORS	PAPI	PAPI Unknown Errors	NO ERRORS	Pubsub	PubSub Unknown Errors	NO ERRORS	RJF	RJF Unknown Errors	NO ERRORS	PayrollDM	PayrollDM Unknown Errors	NO ERRORS	01 Nov 2023	DONE	Singhal, Vikas
App	Dashboard	Status																					
PSP	PSP Unknown Errors	NO ERRORS																					
PAPI	PAPI Unknown Errors	NO ERRORS																					
Pubsub	PubSub Unknown Errors	NO ERRORS																					
RJF	RJF Unknown Errors	NO ERRORS																					
PayrollDM	PayrollDM Unknown Errors	NO ERRORS																					
13	Meet to decide GO/No GO	Decide if we are good to go for Cutover	30 Oct 2023	DONE	N, Ramasubramani Muthurangam, KarthikeyanNasim, Syed Iqbal Choubey, Mayank Bhat, Chethan																		
14	Send Downtime Update to stake holders via Email.	Refer NotificationEmailAddresses Section. Ask the stake holders to be prepared for the downtime.	31 Oct 2023	DONE	Bhat, Chethan																		
15	Reassign PagerDuty.	1. Update PagerDuty Schedule for the Cutover window. 2. Update PagerDuty Schedule for the next week after the Cutover. Pager Duty Schedules QBPY - DPS : EMS OPS RNO Primary SBG - QBPY - DPS : DevOps PSP (IDC)	02 Nov 2023	DONE	Bhat, Chethan																		
16	Note down all the PSP JSS Jobs in the same window.	Link : PSP JSS View Update Failed/MissedBatchJobs Section.	04 Sept 2023	DONE	Karra, Vineeth Hoolimath, Praveenkumar																		
17	Note down all the RJF JSS Jobs in the same window.	Link : RJF JSS View Update Failed/MissedBatchJobs Section.	04 Sept 2023	DONE	Hoolimath, Praveenkumar Karra, Vineeth																		
18	IXP flag updates	Update ENABLE_RUNTIME_HINT and RUNTIME_HINT_VALUES IXP flag	02 Nov 2023	DONE	Goswami, Harshita																		
19	Update IDPS Postgres Hash Credentials. 1. dataAccess.connection.url 2. dataAccess.connection.username 3. dataAccess.connection.password 4. dataAccess.connection.readOnly.url 5. dataAccess.connection.readOnly.username 6. dataAccess.connection.readOnly.password	Steps: Pre-requisite: install StashCopy .pem file to /tmp folder. eiamcli login - Add Secrets from Command Line. sh Oracle-Hash-A.sh -a sh Postgres-Hash-B.sh -a sh Oracle-Hash-C.sh -a - Validate Secrets from Command Line. sh Oracle-Hash-A.sh -v sh Postgres-Hash-B.sh -v sh Oracle-Hash-C.sh -v - Validate from Developer Portal	01 Nov 2023	DONE	Bhagat, Suyash																		
20	Update CR and Image Details	Create Change Requests & decide images for Cutover and Rollback. Refer ChangeRequests Section	02 Nov 2023	DONE	Unknown User (mvenkatesh)																		
21	Bring down staging and middleware apps.	- Bring down staging and middleware apps. - Ensure no database connections from Middleware and Staging. - Bringdown the databases.	02 Nov 2023	DONE	Bahl, Pulkit Choubey, Mayank																		
22	Ensure there is no OOS or any data issue between Oracle Hash and Postgres Hash	- Veridata reports should not have any OOS. - Ensure latest/last 9 days and 1 day compare is green. Veridata Console - Job Name: Rajoria, Himanshu PSP_O2P_last_2_days_B_C2 PSP_O2P_last_90_days_B_C2 PSP_O2P_year2023_except_60_days_B_C3	02 Nov 2023	COMPLETED	Rajoria, Himanshu																		

23	Run Sanity for PSP/PAPI/Pubsub/RJF. Execute the sanity at 9AM on the Cutover day.	Execute below Sanity Jobs to ensure jobs working fine as expected. <table><tr><th>App</th><th>Sanity Link</th><th>Status</th></tr><tr><td>PSP</td><td>PSP Sanity</td><td>SUCCESS</td></tr><tr><td>PAPI</td><td>PAPI Sanity</td><td>SUCCESS</td></tr><tr><td>Pubsub</td><td>Pubsub Sanity</td><td>SUCCESS</td></tr><tr><td>RJF</td><td>RJF Sanity</td><td>SUCCESS</td></tr><tr><td>PayrollDM</td><td>PayrollDM Sanity</td><td>SUCCESS</td></tr></table>	App	Sanity Link	Status	PSP	PSP Sanity	SUCCESS	PAPI	PAPI Sanity	SUCCESS	Pubsub	Pubsub Sanity	SUCCESS	RJF	RJF Sanity	SUCCESS	PayrollDM	PayrollDM Sanity	SUCCESS	03 Nov 2023	DONE	Goswami, Harshita
App	Sanity Link	Status																					
PSP	PSP Sanity	SUCCESS																					
PAPI	PAPI Sanity	SUCCESS																					
Pubsub	Pubsub Sanity	SUCCESS																					
RJF	RJF Sanity	SUCCESS																					
PayrollDM	PayrollDM Sanity	SUCCESS																					
24	Validate PSP Node Level Sanity.	Validate Node level Sanity and Dashboard Node Level Sanity Dashboard .	02 Nov 2023	DONE	Goswami, Harshita																		
25	Update Security Group to add Production Connection	Update Security Group to add Production Connection	02 Nov 2023	COMPLETED	Choubey, Mayank																		
26	Integration Test Suite	https://build.intuit.com/intuit-desktop-payroll/job/iks-intg/5020/testReport/  iks-integration test results.pdf	01 Nov 2023	DONE	Unknown User (mvenkatesh)																		
27	Automation Test Suite	https://build.intuit.com/intuit-desktop-payroll/job/payroll-automation-suite-e2e/2673/cucumber-html-reports/overview-features.html  Automation test resport.pdf	01 Nov 2023	DONE	Unknown User (mvenkatesh)																		

28	Team to validate Database connectivity for A/B/C Prod instances.	• Choubey, Mayank to create a read-only user on A/B/C instances. • Team to validate Database connectivity for A/B/C Prod instances.	02 Nov 2023	DONE	Choubey, Mayank Bhagat, Suyash Goswami, Harshita
29	Update Maintenance Window	Update Maintenance Window. Refer Maintenance Windows section. • Friday (11AM - 4PM) IST Refer below threads and confirm they snoozed their asset as well. • mse-signoff • cx3-production-signoff	02 Nov 2023	DONE	Bhagat, Suyash
30	Create PR to update Config Map in master deployment branch and update the pull request section		02 Nov 2023	NA	Singh, Sanket
31	Run/Pause the Jobs	• Run/Pause the LedgerBalance Job after 12:30PM IST • Run/Pause the State report processor after 12:30PM IST • Pause SoxDBReport • Pause RTBAutomationJob	03 Nov 2023	OPEN	Hoolimath, Praveenkumar

Cutover Steps

S No	Task	Steps	Status	Owner	Remarks
1	Update Maintenance Window	Update Maintenance Window. Refer Maintenance Windows section. • Friday (11AM - 4PM) IST Refer below threads and confirm they snoozed their asset as well. • MSE Team • CX3 Team	OPEN	Bhagat, Suyash Singh, Sanket	
2	Data Parity	Ensure latest 1 day compare and sanity check output is green.	OPEN	Rajoria, Himanshu	
3	Send Downtime Update to stake holders via Email.	CR, Apps & Downtime details to be shared. Refer ChangeRequests Section. Refer NotificationEmailAddresses Section.	OPEN	Bhat, Chethan	
4	Specific communication to #ioc Nasim, Syed Iqbal	Inform #ioc team on ICP services not being snoozed and leading and it leading to incident. This is required because of Tax HCAP and ICP team dont want to miss the alerts.	OPEN	Nasim, Syed Iqbal	
5	Inform partner teams slack channel.	Ask the stake holders to be prepared for the downtime. CR, Apps & Downtime details to be shared. Refer ChangeRequests Section. Inform following Slack Channels. Partner teams communication • ioc-signoff • sbg-payroll-prod-signoff • psp-rtb-priority-signoff • payroll-pd-releases-signoff • psp-inform • mse-signoff • cx3-production-signoff	OPEN	Bhat, Chethan	
6	Ensure there are no critical jobs running.	• Navigate to JSS View Link. • Wait for critical jobs to complete.	OPEN	Hoolimath, Praveenkumar Karra, Vineeth	

7	Bring down below apps. 1. PayrollDM 2. Pubsub 3. PAPI 4. RJF	Bring down Pubsub, PAPI, RJF and PayrollDM. Zero Pods With Maintenance Page	OPEN	Joshi, Vishwanath																						
8	Bring down PSP along with Maintenance Page	Bring down PSP with Maintenance Page. Zero Pods PSP With Maintenance Page	OPEN	Joshi, Vishwanath																						
9	Ensure all apps are down.	Hit following URLs for each application. <table><tr><th>App</th><th>Health Check URL</th><th>Status</th></tr><tr><td>PSP</td><td>sh PSPHealthCheck.sh prd</td><td> SUCCESS </td></tr><tr><td>SAP</td><td>SAP</td><td> SUCCESS </td></tr><tr><td>PAPI</td><td>PAPI Health Check</td><td> SUCCESS </td></tr><tr><td>Pubsub</td><td>Pubsub Sanity</td><td> SUCCESS </td></tr><tr><td>RJF</td><td>RJF Health Check</td><td> SUCCESS </td></tr><tr><td>PayrollDM</td><td>PayrollDM Health Check</td><td> SUCCESS </td></tr></table>	App	Health Check URL	Status	PSP	sh PSPHealthCheck.sh prd	SUCCESS	SAP	SAP	SUCCESS	PAPI	PAPI Health Check	SUCCESS	Pubsub	Pubsub Sanity	SUCCESS	RJF	RJF Health Check	SUCCESS	PayrollDM	PayrollDM Health Check	SUCCESS	OPEN	Rajvanshi , Venkat Singh, Sanket	
App	Health Check URL	Status																								
PSP	sh PSPHealthCheck.sh prd	SUCCESS																								
SAP	SAP	SUCCESS																								
PAPI	PAPI Health Check	SUCCESS																								
Pubsub	Pubsub Sanity	SUCCESS																								
RJF	RJF Health Check	SUCCESS																								
PayrollDM	PayrollDM Health Check	SUCCESS																								
10	Update the downtime details section for all the apps.	Update DowntimeDetails Section.	OPEN	Bhat, Chethan																						
11	Note down All the failed RJF/PSP BatchJobs.	Update Failed/MissedBatchJobs Section.	OPEN	Hoolimath, Praveenumar Karra, Vineeth																						
12	Lock application users on Oracle Hash db.	- Wait for Inflight requests/Running Batch Jobs to complete. - There should not be any connections to Oracle Hash Database. - Lock Oracle Monolith Users : PSPADM and PSPAPP <pre>alter user PSPADM account lock; alter user PSPAPP account lock; alter user PSPREAD account lock; alter user PSPREAD_AWS account lock; select username,account_status from dba_users where username in ('PSPAPP','PSPADM','PSPREAD','PSPREAD_AWS');</pre>	OPEN	Choubey, Mayank																						
13	Reboot readers	psp-prod-uw022 psp-prod-uw023 psp-prod-uw024	OPEN	Choubey, Mayank																						

14	Ensure there are no active connections to the pre-production Oracle Hash database.	No active connections from PSP / PAPI / Pubsub / RJF. Monolith and Monolith ReadReplica : <pre> set lines 300 col username for a30 col machine form a70; select username,machine,count(*) from gv\$session where username is not NULL and username not in ('RDSADMIN','INTUADMIN','SYS','SYSTEM') and type = 'USER' group by service_name, username,machine order by username,machine; </pre> (Kill Connections if needed) Monolith <pre> select 'exec rdsadmin.rdsadmin_util.kill ' s.sid ',' s. serial# ',' ''PROCESS'' ');' from gv\$session s where USERNAME in ('PSPAPP','PSPADM'); </pre>	OPEN	Choubey, Mayank	
15	Check replication lag	check replication lag for C2->D2, C2D2'	OPEN	Choubey, Mayank	
16	Ensure replication lag is zero	Wait till Oracle Hash (B) Postgres Hash(C2) Replication lag to zero.	OPEN	Choubey, Mayank	
17	Create and alter sequences	Alter sequences on C2 to start with last_number from oracle +1 <pre> SELECT 'ALTER SEQUENCE ' sequence_owner '.' sequence_name ' RESTART WITH ' (last_number + 1) ';' FROM all_sequences WHERE sequence_owner = 'PSPADM'; </pre> ggpspadm01.sbg-ppsp-prod.a.intuit.com <pre> cd /u01/switchover_to_C2 --Get Seq val from B @Sequence_Select_From_B.sql --Sequeunce Reset on C2 \i seq_alter_seq_onC2.sql </pre>	OPEN	Choubey, Mayank	
18	Enable Triggers	Enable Triggers on C2. login to C2 host : ggpspadm01.sbg-ppsp-prod.a.intuit.com <pre> cd /u01/switchover_to_C2 \i enable_Triggers_on_C2.sql </pre>	OPEN	Choubey, Mayank	

19	Disable FK Constraints & triggers on Oracle Hash	Disable all FKs & triggers on B after setting settag=2 host : ggpspadm01.sbg-ppsp-prod.a.intuit.com <pre>@disable_triggers_on_B.sql @disable_constraints_on_B.sql</pre> Validation count is 0 for both of below <pre>select constraint_name, table_name,status from dba_constraints where owner='PSPADM' and constraint_type='R' and status='ENABLED';</pre> <pre>select trigger_name,table_name,status from dba_triggers where owner='PSPADM' and status='ENABLED';</pre>	OPEN		
20	Create & Start Extract on C2	Hostname - ggpspagg01.sbg-ppsp-prod.a.intuit.com Register Extract <pre>obey ./diroby/dblogin_c2.oby register extract EPP1PG02 with database pspagg02</pre> Add Extract <pre>obey ./diroby/create_epp1pg02.oby start EPP1PG02</pre> Test with insert on heartbeat table to ensure its extracting data from C2 <pre>insert into pspadm.gg_heartbeat values ('C2_to_B_03Nov',current_timestamp); stats EPP1PG02</pre>	OPEN	Choubey, Mayank	
21	Create pump for C2 to B	Hostname - ggpspagg01.sbg-ppsp-prod.a.intuit.com <pre>obey ./diroby/create_ppsphp02.oby info ppsphp02 start ppsphp02 stats ppsphp02</pre>	OPEN	Choubey, Mayank	

22	Create & Start Replicat for C2->B.	Create & Start Replicat for C2->B. (settag=2, and during downtime) Hostname - ggpsphp05.sbg-ppsp-prod.a.intuit.com Add the Relicats <pre>obey ./diroby/create_replicats.oby obey ./diroby/create_coordinated_replicats.oby</pre> Register the Replicats <pre>dblogin userdalias ggtarget register replicat rpspg51 database register replicat rpspg52 database register replicat rpspg53 database register replicat rpspg54 database register replicat rpspg55 database register replicat rpspg56 database register replicat rpspg57 database register replicat rpspg58 database</pre>	OPEN	Choubey, Mayank	
23	Validate replication C2->B using heartbeat table	Validate replication C2->B using heartbeat table Validate replication C2->D2 using heartbeat table <pre>For C2->B start RPPG6 stats RPPG6 For C2->D2 stats RPPG3 select * from pspadm.gg_heartbeat;</pre>	OPEN	Choubey, Mayank	
24	Unlock users in Postgres Hash (C2)	<pre>alter user pspapp with login; alter user pspread with login; alter user pspadm_owner with login; alter user pspbatch_ro_user with login; alter user pspbatch_rw_user with login; select rolname,rolcanlogin from pg_roles where rolname like '%psp%' order by rolname;</pre>	OPEN	Choubey, Mayank	
25	Restart extract on B (for B->C2)	Hostname - ggpsphp05.sbg-ppsp-prod.a.intuit.com Edit extract parameter file to include TRANLOGOPTIONS FETCHPARTIALLOB <pre>stop EPSPHNP2 edit parameter file epsphpn2.prm start EPSPHNP2</pre>	OPEN	Choubey, Mayank	

26	Validate Production to point to Postgres Hash(C2) Database.	Validate Read Write / VMP / RJF Read Instances (3 instances in total). • Pre-requisite: install Stash • Copy .pem file to /tmp folder. • elamcli login • Validate IDPS Credentials (Database URL/Username/Password) <i>sh Postgres-Hash-B.sh -v</i>	OPEN	Bhagat, Suyash										
27	Bring up PSP Pods with Maintenance.	Bring up PSP with Maintenance page. Scale up PSP Pods With Maintenance Page. Ensure correct image is deployed. • Dashboard : Validate from DT Payroll Pod Info • Script : imageDetails.sh prd	OPEN	Joshi, Vishwanath										
28	Monitor if PSP connecting to Postgres Database	Monitor Database Connection Dashboard and validate.	OPEN	Bahl, Pulkit										
29	Ensure there are no errors in the startup logs.	Look for errors in this dashboard Unknown Errors	OPEN	Bhagat, Suyash										
30	Check the Node level Sanity Logs.	Node Level Sanity Dashboard - Check if all the pods passed sanity tests.	OPEN	Rajvanshi, Venkat										
31	Monitor DB Connections.	Monitor db connetions	OPEN	Choubey, Mayank										
32	Monitor Overall database/replication health vi CW dashboard	Postgres Database Performance Monitor GoldenGate Replication Dashboard	OPEN	Choubey, Mayank										
33	Validate Config Map in PSP Argo, monolith should be pointing to Postgres.	Production PSP ArgoCd	OPEN	.. Prince										
34	Bring up PSP pods without Maintenance.	Bring up PSP without Maintenance page. Scale up PSP Pods Without Maintenance Page. Ensure correct image is deployed. • Dashboard : Validate from DT Payroll Pod Info • Script : imageDetails.sh prd	OPEN	Joshi, Vishwanath										
35	Run the PSP Sanity.	Execute below Sanity Jobs to ensure jobs working fine as expected. <table><tr><th>Application</th><th>Sanity Link</th><th>Status</th></tr><tr><td>PSP</td><td>PSP Sanity</td><td> OPEN </td></tr><tr><td>SAP</td><td>SAP Sanity</td><td> OPEN </td></tr></table>	Application	Sanity Link	Status	PSP	PSP Sanity	OPEN	SAP	SAP Sanity	OPEN	OPEN	Singhal, Vikas	
Application	Sanity Link	Status												
PSP	PSP Sanity	OPEN												
SAP	SAP Sanity	OPEN												
36	Update the downtime details section for PSP.	Update DowntimeDetails Section.	OPEN	Bhat, Chethan										
37	Monitor DB Performance	We will be using combination of Performance Insights, monitoring dashboard and adhoc scripts / slow query log based on need	OPEN	Choubey, Mayank Popat, Ketan Gunvantrai										
38	Bring up all pods for below services 1. PAPI 2. Pubsub 3. PayrollDM 4. RJF	Bring up Pubsub, PAPI, RJF and PayrollDM apps without maintenance page. Bring up Apps Without Maintenance Page Ensure correct image is deployed. • Dashboard : Validate from DT Payroll Pod Info • Script : imageDetails.sh prd	OPEN	Joshi, Vishwanath										

39	Validate Config Map in Argo for the below apps, monolith should be pointing to Postgres. 1. PAPI 2. Pubsub 3. PayrollDM	<table><tr><th>Application</th><th>Argo Link</th></tr><tr><td>PAPI</td><td>Production PAPI ArgoCd</td></tr><tr><td>Pubsub</td><td>Production Pubsub ArgoCd</td></tr><tr><td>RJF</td><td>Production RJF ArgoCd</td></tr><tr><td>PayrollDM</td><td>Production PayrollDM ArgoCd</td></tr></table>	Application	Argo Link	PAPI	Production PAPI ArgoCd	Pubsub	Production Pubsub ArgoCd	RJF	Production RJF ArgoCd	PayrollDM	Production PayrollDM ArgoCd	OPEN	.., Prince						
Application	Argo Link																			
PAPI	Production PAPI ArgoCd																			
Pubsub	Production Pubsub ArgoCd																			
RJF	Production RJF ArgoCd																			
PayrollDM	Production PayrollDM ArgoCd																			
40	Monitor if all apps connecting to Postgres Database	Monitor Database Connection Dashboard and validate.	OPEN	Bahl, Pulkit																
41	Ensure there are no errors in the startup logs.	Monitor and Report unknown errors. <table><tr><th>Application</th><th>Dashboard</th></tr><tr><td>PAPI</td><td>PAPI Unknown Errors</td></tr><tr><td>Pubsub</td><td>PubSub Unknown Errors</td></tr><tr><td>RJF</td><td>RJF Unknown Errors</td></tr><tr><td>PayrollDM</td><td>Payroll DM Unknown Errors</td></tr></table>	Application	Dashboard	PAPI	PAPI Unknown Errors	Pubsub	PubSub Unknown Errors	RJF	RJF Unknown Errors	PayrollDM	Payroll DM Unknown Errors	OPEN	Singhal, Vikas						
Application	Dashboard																			
PAPI	PAPI Unknown Errors																			
Pubsub	PubSub Unknown Errors																			
RJF	RJF Unknown Errors																			
PayrollDM	Payroll DM Unknown Errors																			
42	Run PAPI/Pubsub/PayrollDM/RJF Sanity.	Execute below Sanity Suites. <table><tr><th>Application</th><th>Sanity Link</th><th>Status</th></tr><tr><td>PAPI</td><td>PAPI Sanity</td><td> OPEN </td></tr><tr><td>Pubsub</td><td>Pubsub Sanity</td><td> OPEN </td></tr><tr><td>RJF</td><td>RJF Sanity</td><td> OPEN </td></tr><tr><td>PayrollDM</td><td>PayrollDM Sanity</td><td> OPEN </td></tr></table>	Application	Sanity Link	Status	PAPI	PAPI Sanity	OPEN	Pubsub	Pubsub Sanity	OPEN	RJF	RJF Sanity	OPEN	PayrollDM	PayrollDM Sanity	OPEN	OPEN	Rajvanshi , Venkat Singh, Sanket	
Application	Sanity Link	Status																		
PAPI	PAPI Sanity	OPEN																		
Pubsub	Pubsub Sanity	OPEN																		
RJF	RJF Sanity	OPEN																		
PayrollDM	PayrollDM Sanity	OPEN																		
43	Ensure we see DB Connections from Pubsub / PAPI / PayrollDM /RJF.	Monitor db connetions	OPEN	Choubey, Mayank																
44	Update the downtime details section for all the apps.	Update DowntimeDetails Section.	OPEN	Bhat, Chethan																
45	Run PSP & RJF failed jobs and missed Jobs.	Refer Failed/MissedBatchJobs Section. Run LedgerBalance Job if Missed.	OPEN	Hoolimat h, Praveenk umar Karra, Vineeth																
46	Monitor all the apps and databases.	Refer Monitoring Section.	OPEN	Everyone																
47	Ensure replication is running without any issues & monitor the lag	GoldenGate Replication Dashboard	OPEN	Choubey, Mayank																
48	Review & Take action on all the pending Alerts.	Pending Alerts - Critical Alerts PD Pending Alerts - RNO PD	OPEN	N, Ramasub ramani / Bhat, Chethan																
49	Run Payroll From Production Test company	IKS QBDT PSP Adapters	OPEN	K, Srividya																
50	Monitor RJF & PSP Batch Jobs.	- Update JSS Jobs Schedule completed jobs status every 2hours. (validate job using sql queries) - Update RJF Jobs Schedule completed jobs status every 2hours. (validate job with splunk)	OPEN	Hoolimat h, Praveenk umar Karra, Vineeth																
51	Merge generated PRs to environment/prd-usw2-iks to update config map in manifest to point to postgres for PSP Papi Pubsub		OPEN	Bhagat, Suyash																

52	Merge PRs to the master branch to update config map to postgres for PSP Papi Pubsub refer PR section		NA	.. Prince	
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Post Cutover Steps

S No	Task	Steps	Status	Owner	Remarks
1	Run Ledger Balance & State Report job if required.	Run Ledger Balance & State Report job if required.	OPEN	Bansal, Jatin	
2	Send Downtime Update to stake holders via Email.	Refer NotificationEmailAddresses Section.	OPEN	Bhat, Chethan	
3	Offline Monitoring for all the Apps.	Refer Monitoring Section. Update once in every 2 hours in Cutover slack channel.	OPEN	All	
4	Review & Take action on all the pending Alerts.	Pending Alerts - Critical Alerts PD Pending Alerts - RNO PD	OPEN	Hoolimath, Praveenkumar Karra, Vineeth	
5	Review & Close all pending CRs.	Refer ChangeRequests Section.	OPEN		
6	Inform partner teams slack channel.	Inform following Slack Channels. Partner teams communication ioc-signoff sbg-payroll-prod-signoff psp-rtb-priority-signoff payroll-pd-releases-signoff psp-inform mse-signoff cx3-production-signoff	OPEN	Bhat, Chethan	
7	Monitor Critical Batch Jobs.	Monitor Critical Batch Jobs (Required Only for the Cutover) Refer CriticalBatchJobsMonitoring Section.	OPEN	Hoolimath, Praveenkumar	
8	Monitor database Stability, Availability & Performance	We will be using combination of Performance Insights, monitoring dashboard and adhoc scripts / slow query log based on need	OPEN	Choubey, Mayank (Primary) Popat, Ketan Gunvantrai (Escalation)	
9	Monitor replication from Postgres to Oracle Hash	- Monitor DMS replication lag from Postgres to Oracle Hash - Keep running sanity counts. - Ensure Veridata validation is green from Postgres to Oracle Hash	OPEN	Narlagalla, Praveen Goud	
10	Schedule Veridata to validate Delta Data.	Schedule Veridata to validate Delta Data. Veridata Console Link Veridata Job Name: Rajoria, Himanshu PSP_P2O_last_6_hrs_C2_B	COMPLETED	Rajoria, Himanshu	
11	Apply Repair Scripts if applicable.	Apply Repair Scripts if applicable.	OPEN	Choubey, Mayank	

Emergency Rollback Decision and Action Plan

Emergency Rollback and FMEA decisions documented [here](#)

Emergency Rollback Steps

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1	Update Maintenance Window	Update Maintenance Window. Refer Maintenance Windows section. - Sunday (10AM - 1PM) IST - Monday (10AM - 1PM) IST Refer below threads and confirm they snoozed their asset as well. MSE Team CX3 Team	OPEN	Singh, Sanket															
2	Monitor Postgres Hash --> Oracle Hash Replication lag	Monitor C2B, C2D2, C2->D2 Replication lag and ensure on lag.	OPEN	Unknown User (rrasam) Rajoria, Himanshu Choubey, Mayank															
3	Data Parity - Ensure Veridata Validation on Latest Data is complete.	Trigger validation for the Tip-toe period for C2B, C2D2, C2->D2	OPEN																
4	Note down all the PSP JSS Jobs in the same window.	Link : PSP JSS View Update Failed/MissedBatchJobs Section.	OPEN	Karra, Vineeth															
5	Note down all the RJF JSS Jobs in the same window.	Link : RJF JSS View Update Failed/MissedBatchJobs Section.	OPEN	Karra, Vineeth															
6	Inform slack channel about the downtime.	Ask the stake holders to be prepared for the downtime. CR, Apps & Downtime details to be shared. Refer ChangeRequests Section.Inform following Slack Channels. #payroll-pd-releases, #ioc, #psp-rtb-priority, #sbg-payroll-prod, #mse-commerce-support, #cx3-production payroll-pd-releases ioc psp-rtb-priority sbg-payroll-prod mse-commerce-support cx3-production	OPEN	Bhat, Chethan															
7	Send Downtime Update to stake holders via Email.	CR, Apps & Downtime details to be shared. Refer ChangeRequests Section. Refer NotificationEmailAddresses Section.	OPEN	Bhat, Chethan															
8	Ensure there are no critical batch jobs /RJF jobs running.	Refer JSS View Link. Wait for critical jobs to complete.	OPEN	Karra, Vineeth															
9	Bring down below pods 1. Pubsub 2. RJF 3. PAPI 4. PayrollDM	Bring down Pubsub, PAPI (With Maintenance Page) and RJF. Zero Pods With Maintenance Page	OPEN	Joshi, Vishwanath															
10	Bring down PSP along with Maintenance Page	Bring Down PSP with Maintenance Page. Zero Pods PSP With Maintenance Page.	OPEN	Joshi, Vishwanath															
11	Ensure all apps are down.	Hit Following URLs and make sure all apps are down. <table><tr><th>Application</th><th>Health Check</th></tr><tr><td>PSP</td><td>sh PSPHealthCheck.sh prd</td></tr><tr><td>SAP</td><td>SAP</td></tr><tr><td>PAPI</td><td>PAPI Health Check</td></tr><tr><td>Pubsub</td><td>Pubsub Sanity</td></tr><tr><td>RJF</td><td>RJF Health Check</td></tr><tr><td>PayrollDM</td><td>PayrollDM Health Check</td></tr></table>	Application	Health Check	PSP	sh PSPHealthCheck.sh prd	SAP	SAP	PAPI	PAPI Health Check	Pubsub	Pubsub Sanity	RJF	RJF Health Check	PayrollDM	PayrollDM Health Check	OPEN	Singh, Sanket	
Application	Health Check																		
PSP	sh PSPHealthCheck.sh prd																		
SAP	SAP																		
PAPI	PAPI Health Check																		
Pubsub	Pubsub Sanity																		
RJF	RJF Health Check																		
PayrollDM	PayrollDM Health Check																		
12	Update the downtime details section for all the apps.	Update DowntimeDetails Section.	OPEN	Bhat, Chethan															
13	Note down All the failed RJF/PSP BatchJobs.	Update Failed/MissedBatchJobs Section.	OPEN	Karra, Vineeth															

14	Lock application users on Postgres Hash db.	• Wait for Inflight requests/Running Batch Jobs to complete. • There should not be any connections to Postgres Hash Database. • Lock Postgres Hash Users. <pre>alter user pspapp with nologin; alter user pspadm_owner with nologin; alter user pspbatch_ro_user with nologin; alter user pspbatch_rw_user with nologin; alter user psprjf with nologin;</pre>	OPEN	Choubey, Mayank	
15	Ensure there are no active connections to the Postgres Hash Database.	No active connections from PSP / PAPI / Pubsub / RJF / PayrollDM. Monolith : <pre>SELECT datname,username, coalesce (client_hostname, client_addr::text, 'localhost') as client_hostname,count(*) from pg_stat_activity where pid != pg_backend_pid() group by datname,username, coalesce (client_hostname, client_addr::text, 'localhost') order by username;</pre> (Kill Connections if needed) Monolith <pre>select pg_terminate_backend(pid) from pg_stat_activity where pid in (select pid FROM pg_stat_activity where username like '%psp%');</pre>	OPEN	Choubey, Mayank	
16	Monitor Postgres Hash Oracle Hash Replication Lag	Wait till Postgres Hash(C2) --> Oracle Hash(B), C2D2, C2D2' Replication lag to zero.	OPEN	Choubey, Mayank	
17	GG Replication from Oracle Hash(B) Postgres Hash(C2)	Ensure Extract (B) (for B->C2) stats should not have any table extract <pre>stats EPSHPNP2</pre>	OPEN	Choubey, Mayank	
18	Disable triggers on C2	Host: ggpspadm01.sbg-ppsp-prod.a.intuit.com <pre>\i disable_triggers_on_C2.sql</pre> Validate triggers are disabled.	OPEN	Narlagalla , Praveen Goud	
19	alter sequences start value on B	Alter sequences to start with last_number from C2 +1 ggpspadm01.sbg-ppsp-prod.a.intuit.com <pre>\i seq_alter_seq_on_B.sql</pre>	OPEN	Narlagalla , Praveen Goud	

20	Enable all triggers, FKs on B	Enable all triggers, FKs on B using SETTAG 2 (to exclude going to C2) Validation <pre>select constraint_name, table_name,status from dba_constraints where owner='PSPADM' and constraint_type='R' and status='DISABLED';</pre> <pre>select trigger_name,table_name,status from dba_triggers where owner='PSPADM' and status='DISABLED';</pre>	OPEN	Choubey, Mayank	
21	Test data extraction from B Extract. Not required as gg_heartbeat job is running on B	Test with insert on heartbeat table to ensure its extracting data from B <pre>insert into pspadm.gg_heartbeat values ('B_to_C2', sysdate);</pre> <pre>stats EPSHPNP2</pre>	OPEN	Choubey, Mayank	
22	Unlock Oracle Hash Users	Unlock Oracle Hash Users. <pre>alter user PSPADM account unlock; alter user PSPAPP account unlock; alter user PSPREAD account unlock; alter user PSPREAD_AWS account unlock; select username,account_status from dba_users where username in ('PSPAPP','PSPADM','PSPREAD','PSPREAD_AWS');</pre>	OPEN	Choubey, Mayank	
23	Cleanup GG processes for C2->B	Delete extract on C2 <pre>obey ./diroby/dblogin_c2.oby info EPP1PG02 detail stop EPP1PG02 delete EPP1PG02 UNREGISTER EXTRACT EPP1PG02 with database pspapg02</pre> Delete pump on C2 <pre>info ppsphp02 detail stop ppsphp02 delete extract ppsphp02</pre> Delete replicat on B <pre>obey ./diroby/delete_replicats.oby</pre>	OPEN	Choubey, Mayank	
24	Validate Production to point to Oracle Hash(A) Database.	Validate Read Write / VMP / RJF Read Instances (3 instances in total). Pre-requisite: install Stash - Copy .pem file to /tmp folder. - eiampcli login - Validate IDPS Credentials (Database URL/Username/Password) <pre>sh Oracle-Hash-A.sh -v</pre>	OPEN	Bhagat, Suyash	

25	Bring up PSP pods with Maintenance on Oracle	Bring up PSP with Maintenance page. Scale up PSP Pods With Maintenance Page. Ensure correct image is deployed. • Dashboard : Validate from DT Payroll Pod Info • Script : imageDetails.sh prd • Validate database in config map	OPEN	Joshi, Vishwanath											
26	Monitor if PSP connecting to Oracle Hash Database	Monitor Database Connection Dashboard and validate.	OPEN	Bahl, Pulkit											
27	Ensure there are no errors in the startup logs.	Look for errors in this dashboard Unknown Errors	OPEN	Goswami, Harshita											
28	Check the Node level Sanity Logs.	Node Level Sanity Dashboard - Check if all the pods passed sanity tests.	OPEN	Bahl, Pulkit											
29	Monitor DB Connections on Oracle Hash.	Monitor db connetions	OPEN	Choubey, Mayank											
30	Monitor Overall database/replication health vi CW dashboard	Oracle Hash Database Performance Monitor GoldenGate Replication Dashboard	OPEN	Choubey, Mayank											
31	Validate Config Map in PSP Argo , monolith should be pointing to Oracle Hash.	Production PSP ArgoCd	OPEN	@Prince											
32	Bring up PSP pods without Maintenance on Oracle Hash.	Bring up PSP without Maintenance page. Scale up PSP Pods Without Maintenance Page. Ensure correct image is deployed. • Dashboard : Validate from DT Payroll Pod Info • Script : imageDetails.sh prd • Validate database in config map	OPEN	Joshi, Vishwanath											
33	Run the PSP Sanity.	Execute below Sanity Jobs to ensure jobs working fine as expected. <table><tr><th>Application</th><th>Sanity Link</th><th>Status</th></tr><tr><td>PSP</td><td>PSP Sanity</td><td> OPEN </td></tr><tr><td>SAP</td><td>SAP Sanity</td><td> OPEN </td></tr></table>	Application	Sanity Link	Status	PSP	PSP Sanity	OPEN	SAP	SAP Sanity	OPEN	OPEN	Singh, Sanket		
Application	Sanity Link	Status													
PSP	PSP Sanity	OPEN													
SAP	SAP Sanity	OPEN													
34	Monitor DB Performance	We will be using combination of Performance Insights, monitoring dashboard and adhoc scripts / slow query log based on need	OPEN	Choubey, Mayank / Popat, Ketan Gunvantrai											
35	Validate Config Map in Argo for the below apps, monolith should be pointing to Oracle Hash. 1. PAPI 2. RJF 3. Pubsub 4. PayrollIDM	<table><tr><th>Application</th><th>Argo Link</th></tr><tr><td>PAPI</td><td>Production PAPI ArgoCd</td></tr><tr><td>Pubsub</td><td>Production Pubsub ArgoCd</td></tr><tr><td>RJF</td><td>Production RJF ArgoCd</td></tr><tr><td>PayrollIDM</td><td>Production PayrollIDM ArgoCd</td></tr></table>	Application	Argo Link	PAPI	Production PAPI ArgoCd	Pubsub	Production Pubsub ArgoCd	RJF	Production RJF ArgoCd	PayrollIDM	Production PayrollIDM ArgoCd			
Application	Argo Link														
PAPI	Production PAPI ArgoCd														
Pubsub	Production Pubsub ArgoCd														
RJF	Production RJF ArgoCd														
PayrollIDM	Production PayrollIDM ArgoCd														
36	Bring up all pods for below services on Oracle 1. PAPI 2. RJF 3. Pubsub 4. PayrollIDM	Bring up Pubsub, PAPI and RJF apps without maintenance page. Bring up Apps Without Maintenance Page Ensure correct image is deployed. • Dashboard : Validate from DT Payroll Pod Info • Script : imageDetails.sh prd • Validate database in config map	OPEN	Joshi, Vishwanath											
37	Monitor if all apps connecting to Oracle Hash Database	Monitor Database Connection Dashboard and validate.	OPEN	Bahl, Pulkit											

38	Ensure there are no errors in the startup logs.	Monitor and Report unknown errors. <table><tr><th>Application</th><th>Dashboard</th><th>Status</th></tr><tr><td>PAPI</td><td>PAPI Unknown Errors</td><td> OPEN </td></tr><tr><td>Pubsub</td><td>PubSub Unknown Errors</td><td> OPEN </td></tr><tr><td>RJF</td><td>RJF Unknown Errors</td><td> OPEN </td></tr><tr><td>PayrollDM</td><td>PayrollDM Unknown Errors</td><td> OPEN </td></tr></table>	Application	Dashboard	Status	PAPI	PAPI Unknown Errors	OPEN	Pubsub	PubSub Unknown Errors	OPEN	RJF	RJF Unknown Errors	OPEN	PayrollDM	PayrollDM Unknown Errors	OPEN	OPEN	Goswami, Harshita	
Application	Dashboard	Status																		
PAPI	PAPI Unknown Errors	OPEN																		
Pubsub	PubSub Unknown Errors	OPEN																		
RJF	RJF Unknown Errors	OPEN																		
PayrollDM	PayrollDM Unknown Errors	OPEN																		
39	Run PAPI / Pubsub / RJF / PayrollDM Sanity.	Execute below sanity suites. <table><tr><th>Application</th><th>Sanity Link</th><th>Status</th></tr><tr><td>PAPI</td><td>PAPI Sanity</td><td> OPEN </td></tr><tr><td>Pubsub</td><td>Pubsub Sanity</td><td> OPEN </td></tr><tr><td>RJF</td><td>RJF Sanity</td><td> OPEN </td></tr><tr><td>PayrollDM</td><td>PayrollDM Sanity</td><td> OPEN </td></tr></table>	Application	Sanity Link	Status	PAPI	PAPI Sanity	OPEN	Pubsub	Pubsub Sanity	OPEN	RJF	RJF Sanity	OPEN	PayrollDM	PayrollDM Sanity	OPEN	OPEN	Singh, Sanket	
Application	Sanity Link	Status																		
PAPI	PAPI Sanity	OPEN																		
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RJF	RJF Sanity	OPEN																		
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40	Run PSP & RJF failed jobs and missed Jobs.	Refer Failed/MissedBatchJobs Section. Run LedgerBalance Job if Missed.	OPEN	Karra, Vineeth																
41	Monitor all the apps and databases.	Refer Monitoring Section.	OPEN	All																
42	Ensure replication is running without any issues & monitor the lag	Monitor Replication Lag.	OPEN	Narlagalla , Praveen Goud																
43	Run Payroll From Production Test company	IKS QBDT PSP Adapters	OPEN	Swain, Amrit																
44	Merge generated PRs to environment /prd-usw2-iks to update config map in manifest to point to oracle for PSP Papi Pubsub RJF		OPEN	Bhagat, Suyash																
45	Merge PRs to the master branch to update config map to oracle PSP Papi Pubsub RJF refer PR section		OPEN	Bhagat, Suyash																

Post Rollback Steps

S No	Task	Steps	Status	Owner	Remarks
1	Send Downtime Update to stake holders via Email.	Refer NotificationEmailAddresses Section.	OPEN	N, Ramasubramani	
2	Offline Monitoring for all Apps for 2 hours.	Refer Monitoring Section.	OPEN	All	
3	Review & Take action on all the pending Alerts.	Pending Alerts - Critical Alerts PD Pending Alerts - RNO PD	OPEN	Kaura, Varun	
4	Review & Close all pending CRs.	Refer ChangeRequests Section.	OPEN	N, Ramasubramani	

5	Inform partner teams slack channel.	Ask the stake holders to be prepared for the downtime. Inform following Slack Channels. Partner teams communication • ioc-signoff • sbg-payroll-prod-signoff • psp-rtb-priority-signoff • payroll-pd-releases-signoff • psp-inform • mse-signoff • cx3-production-signoff	OPEN	N, Ramasubramani	
7	Monitor Critical Batch Jobs.	Monitor Critical Batch Jobs (Required Only for the Cutover) Refer CriticalBatchJobsMonitoring Section.	OPEN	Hoolimath, Praveenkumar	
8	Monitor database Stability, Availability & Performance	We will be using combination of Performance Insights, monitoring dashboard and adhoc scripts / slow query log based on need	OPEN	Choubey, Mayank (Primary) Popat, Ketan Gunvantrai (Escalation)	
9	Monitor replication from Oracle Hash to Postgres	• Ensure no replication issue	OPEN	Choubey, Mayank	
10	Trigger Veridata to validate Delta Data.	Trigger Veridata to validate Delta Data. Veridata Console Link : monolith_rhab_incr_1day	OPEN	Unknown User (rrasam) Rajoria, Himanshu	

Monitoring

Apps Health

- Health & Traffic summary shared in the Cutover slack channel.

Application / Database / Infrastructure Category	Owner	Health		
		Previous Week	Cutover	
		27 Oct 2023	03 Nov 2023	04 Nov 2023
QBDT Adapter	Singhal, Vikas	HEALTHY		
QBDTWS Adapter	Singhal, Vikas	HEALTHY		
DIS Adapter	Singhal, Vikas	HEALTHY		
ADE Adapter	Johnson, Anu	HEALTHY		
CDM Adapter	Johnson, Anu	HEALTHY		
SAP Adapter	Johnson, Anu	HEALTHY		
BRM Adapter	Unknown User (akumari3)	HEALTHY		
PTC Adapter	Unknown User (akumari3)	HEALTHY		
Critical Batch Jobs	Bansal, Jatin	HEALTHY		
Batch Jobs during Activity	Bansal, Jatin	HEALTHY		
Infrastructure Monitoring	Hoolimath, Praveenkumar	HEALTHY		
PAPI	Agarwal, Akshit	HEALTHY		
RJF	Hoolimath, Praveenkumar	HEALTHY		
Pubsub	Bhagat, Suyash	HEALTHY		

Payroll DM	Bhagat, Suyash	HEALTHY		
DB Queries Performance	Rajvanshi, Venkat	HEALTHY		
Database	Choubey, Mayank	HEALTHY		
Veridata	Choubey, Mayank	HEALTHY		
ArgoCd	Joshi, Vishwanath	HEALTHY		
Unknown Errors	Bhagat, Suyash	HEALTHY		
Workflows Performance	Rajvanshi, Venkat	HEALTHY		

PSP

S. No	Dashboard	Link	Comments
1	PSP Batch Jobs 1. Successful Jobs 2. Failed Jobs	PSP Batch Jobs	Update JSS Jobs Schedule completed jobs status every 2hours.
2	PSP JSS View 1. Successful Jobs 2. Failed Jobs	PSP JSS View	
3	PSP Service Metrics 1. Response Time 2. No of Requests	PSP Service Metrics	
4	PSP Adapters 1. Response Codes	PSP Adapters	
5	PSP API Gateway 1. TP Numbers (Check - Not Populating) 2. Response Codes	PSP API Gateway	
6	PSP Unknown Exceptions	PSP Unknown Exceptions	
7	Workforce	Workforce	

PAPI

S. No	Dashboard	Link
1	PAPI Gateway 1. TP Numbers (Check - Not Populating) 2. Response Codes	PAPI Gateway
2	PAPI GraphQL Gateway	PAPI GraphQL Gateway

Pubsub

S. No	Dashboard	Link
1	Pubsub (TODO : To enhance to capture all entities)	Pubsub Entity Publish
2	Pubsub TID Analyzer	Pubsub TID Analyzer

Report Job Framework

	Dashboard	Link	Comments
1	RJF JSS View 1. Successful Jobs 2. Failed Jobs	RJF JSS View Splunk Dashboard : QBDT PSP Report Jobs IKS	Update RJF Jobs Schedule completed jobs status every 2hours.

2	Jobs during the Friday window	PSP_COMPANY PSP_STATEID Company_Agency_Info_SUI Cancelled_By_MissedPayrollsBatchJob Company_Agency_Info Company_Agency_Info_SUI_Debit PSP_Entitlement_AppUser PSP_Entitlement_DBUser PSP_contact ASSISTED_DISCO_CUSTOMER_LIST_NEW dd_limit_inf_jirapsp_6681	
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PayrollDM Monitoring

S. No	Dashboard	Link
1	QBDT PSP DG2.0	QBDT PSP DG2.0

Performance

Traffic	Category	Dashboards
Batch	Pubsub Performance	Pubsub Comparison Dashboard
	PayrollDM Performance	PayrollDM Comparison Dashboard
OLTP	Workflow Performance	OLTP Workflow Performance Comparison
	Query Performance	OLTP Query Performance Comparison
	SAP Performance	SAP Query Performance Comparison
	Papi Workflow Performance	Comparison QBDT Papi Metrics Prod
	VMP Workflow Performance	Comparison QBDT VMP Metrics Prod
	PSP Service Metric	PSP Service Metrics
OLAP	Batch Job Performance	Batch Jobs Performance Comparison
	Query Performance	Batch Jobs Query Performance Comparison

Cutover Batch Jobs Monitoring

S No	Jobs	Validation Scripts
1	AMO	
2	Entity Event	
3	EmailGateway	
4	EMSBSToBRMDataSyncProcessor	
5	IAMEmailAddressProcessor	
6	CheckPrint	
7	EntitlementProcessor	
8	QbdtUnprocessedRequestsRetry	
9	WorkersCompProcessor	
10	AssistedUsageDataSyncProcessor	
11	Ledger Balance	

BatchJobs in Friday Window

S. No	Jobs	Comments
1.	EntitlementProcessor LedgerOperationsProcessor AMOMessageProcessor EntityEventProcessor AssistedUsageDataSyncProcessor EmailGatewayProcessor PSPToEMSBSDDataSyncProcessor RetryQbdtUnprocessedRequestProcessor CheckPrintProcessor EftpsResponseProcessor EftpsSendProcessor EDRAssociationFixPISqlJobsProcessor EFTPSOnHoldPaymentPISqlJobsProcessor WorkersCompProcessor EVSCompanyProcessor IamEmailAddressProcessor RealTimeEntityEventRetryProcessor RTBAutomationProcessor EntityInitialLoadProcessor MtlTransactionReportEnrichProcessor BulkWorkforceInviteProcessor CostCoPISqlJobsProcessor PayrollFraudBatchPurgePISqlJobsProcessor EntityEventRetryProcessor AccountServiceSyncExceptionProcessor ACHTraceIdProcessor AMLReportProcessor IndustryReportProcessor LedgerBalanceProcessor NightlyBatchJobsProcessor StateReportProcessor TPSURReportProcessor FsetFilingProcessor	

Critical Batch Jobs Monitoring

S. No	Steps	Jobs
1	Monitor Critical Batch Jobs.	Agarwal, Akshit to update

Oracle Hash Specific Monitoring

S.No	Dashboard	Comments
1	DT Payroll Oracle Hash	
2		
3		

Workflows Monitoring

S. No	Workflow/AppName	Validation Points
1	Run Payroll	Hoolimath, Praveenkumar will run from test company.
2	SAP Monitoring	Singhal, Vikas to update script here
3	QBDT/QBDTWS Monitoring	Swain, Amrit IKS QBDT PSP Adapters

DB Queries Monitoring

S. No	Dashboard	Link
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1	PSP DB Telemetry	Monitor long running queries. PSP DB Telemetry
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Database Monitoring

S.No.	Dashboard	Link
1	Monitor Oracle Hash --> Postgres Replication Lag	GoldenGate Replication Dashboard
2	Ensure Veridata validation is green from Postgres to Oracle Hash	veridata console
3	Monitor Connections and Long Running Queries	
4	Oracle Hash Specific Dashboards	
5	Oracle Hash database dashboard	Oracle Hash Database Performance Monitor

Infrastructure Monitoring

Golden Gate Signals

Application	Module	Golden Gate Signals Link	Owner
PSP	QBDT Adapter	QBDT Adapter Golden Signals	Bansal, Jatin
	ADE Adapter	ADE Adapter Golden Signals	Bansal, Jatin
	BRM Adapter	BRM Adapter Golden Signals	Bansal, Jatin
	DIS Adapter	DIS Adapter Golden Signals	Bansal, Jatin
	EWS Adapter	EWS Adapter Golden Signals	Bansal, Jatin
	PTC Adapter	PTC Adapter Golden Signals	Bansal, Jatin
	CDM Adapter	CDM Adapter Golden Signals	Bansal, Jatin
	Batch Jobs	Batch Jobs Golden Signals	Bansal, Jatin
PAPI	PAPI	PAPI Golden Signals	Rajoria, Himanshu
RJF	RJF	RJF Golden Signals	Joshi, Vishwanath

Wavefront

Application	Wavefront Monitor	Owner
PSP	PSP Wavefront Monitor	Bansal, Jatin
PAPI	PAPI Wavefront Monitor	Rajoria, Himanshu
Pubsub	Pubsub Wavefront Monitor	Agarwal, Akshit
RJF	RJF Wavefront Monitor	Joshi, Vishwanath
PayrollDM	PayrollDM Wavefront Monitor	Agarwal, Akshit

AppDynamics

This is the backup for Infrastructure monitoring.

Environment	App Dynamics Link	Owner
Production	DTPayroll AppDynamics BaseURL	Agarwal, Akshit

Failed/Missed Batch Jobs

Validation queries for the jobs in the Cutover window can be found [Hoolimath](#), [Praveenkumar](#) to update

App	Category	Job Names / Screenshots (Cutover)	Job Names / Screenshots (Rollback)
Batch Jobs	Scheduled jobs	EntitlementProcessor LedgerOperationsProcessor AMOMessageProcessor EntityEventProcessor AssistedUsageDataSyncProcessor EmailGatewayProcessor PSPTToEMSBSDDataSyncProcessor RetryQbdtUnprocessedRequestProcessor CheckPrintProcessor EftpsResponseProcessor EftpsSendProcessor EDRAssociationFixPISqlJobsProcessor EFTPSOnHoldPaymentPISqlJobsProcessor WorkersCompProcessor EVSCompanyProcessor IamEmailAddressProcessor RealTimeEntityEventRetryProcessor RTBAutomationProcessor EntityInitialLoadProcessor MtlTransactionReportEnrichProcessor BulkWorkforceInviteProcessor CostCoPISqlJobsProcessor PayrollFraudBatchPurgePISqlJobsProcessor EntityEventRetryProcessor AccountServiceSyncExceptionProcessor ACHTraceldProcessor AMLReportProcessor IndustryReportProcessor LedgerBalanceProcessor NightlyBatchJobsProcessor StateReportProcessor TPSUReportProcessor FsetFilingProcessor	
	Missed Jobs		
	Failed Jobs		
RJF	Scheduled jobs	PSP_COMPANY PSP_STATEID PSP_Entitlement_AppUser PSP_contact dd_limit_inf_jirapsp_6681	
	Missed Jobs		
	Failed Jobs		